

Saksham Garg

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Atlanta, Georgia

github.com/ATLSaksham10/Portfolio.git

PROFESSIONAL SUMMARY

Aspiring Aerospace & Mechanical Engineer certified in CAD and FAA Part 107 Remote Pilot. Extensive hands-on experience in high-stakes fabrication, including electric vehicles, robotics, and custom drone assembly. Dedicated, precision-oriented builder focused on structural integrity, mechanical systems, and flight-ready hardware. Proven technical problem-solving skills tailored for dynamic shop and field environments. GHP Engineering Alum.

SKILLS

Creo SimScale Ansys OnShape HydroCAD CAD CFD FEA FlashCut CNC Cura LuaScript

FAA Part 107 HecRas CNC Welding (TIG, MIG, Stick) 3D Printing Calibration Lincoln Welding Systems

PCB Soldering Powertools Woodworking Python Java Office Suite

WORK EXPERIENCE

Engineering Intern

Swagelok Georgia, August 2026–Present

- Expanded tubing quality inspection from 50% to 100% of tube types.
- Reconfigured warehouse workflow into a single-line layout, eliminating operator backtracking.
- Building an interactive digital SOP to replace static procedure documents.

Manufacturing Process Engineering Intern

Caterpillar, May 2026 – August 2026

- Designed and deployed a passive roller system in Creo Parametric for an 8-bin wire harness machine to reduce dynamic wire friction during crimping.
- Ran wire bend radius calculations and materials analysis, increasing feed capacity 140% and extending feed component life ~84%.
- Read and produced wire schematics and electrical blueprints for production.

Volunteer Skilled Supervisor

Atlanta Habitat for Humanity, January 2025–Present

- 350+ hours of on-site construction and warehouse operations; recipient of Habitat's Century Club Award.
- 20 homes completed: directed volunteer crews on framing, siding, soffit, decking, cabinetry, and railing installs.

Freelance Photographer & Drone Videographer

Part Time, January 2024–Present

- FAA Part 107 certified remote pilot; plans and flies aerial shoots in line with FAA airspace and safety regulations.
- Captures ground and aerial photo and video coverage for commercial, nonprofit, and cultural events.
- Processes RAW images in Adobe Lightroom and darktable; edits and color-grades video in DaVinci Resolve.

EDUCATION

Senior @ DENMARK HIGH SCHOOL, GEORGIA

Expected graduation May 2027

Dual Enrollment Student @ Georgia Tech

Math Year 1 (1554 Linear Algebra & 2551 MultiVar Calc)

ACCOMPLISHMENTS & AWARDS

- Georgia **Governors Honors Program** Finalist & Attendee 2025 (Engineering)
- Denmark High School Engineering Student of the Year (**Forsyth CTAE Award**)
- Published peer-reviewed research in **AIAA Journal**
- TSA Top 10 @ Nationals, **Winner @ State**, Top 5 in several events @ State
- Finalist @ **GASTC** Device Modification State Competition 2025
- Fermi Honorable Mention @ GT Physics Field Day
- Century Club Award, Habitat for Humanity
- National Merit Scholarship **Semifinalist**
- **FRC Winner** @ District Qualifier, Rookie Award
- UGA Certificate of Merit
- AP Scholar with Distinction
- Most Spirited Player **2** year in a row, boys tennis team
- National Honor Society Member

LEADERSHIP AND COMMUNITY INVOLVEMENT

- President, Co-Captain |** FIRST Robotics Team #11174, Danes Robotics
Founding member, led 20+ student team to build robot and win district qualifier. Worked with CAN PDB.
- Mentor, Officer |** Denmark Technology Student Association
Various officer roles, captained drone team, calibrated 12+ 3D Printers, raised \$2500, grew organization 75%.
- Founder, President |** Enginuity (Community STEM Education Organization)
Designed and 3D Printed low cost, affordable robotics kits. Students assembled them in CAD Workshops.
- AP Ambassador |** Denmark AP Program
Presented Denmark AP Program to over 700+ families, supported incoming students for success in AP
- Student Researcher |** American Institute of Aeronautics & Astronautics
Published research on Urban Air Mobility scaling complexity for future smart cities in AIAA Technical Journal
- CTAE Ambassador |** Forsyth County Schools Superintendent Advisory Committee
Increased vertical integration between schools, introducing pathways and careers to younger students.
- Student Advisor |** Forsyth County Schools Superintendent Advisory Committee
Provided input to Dr. Mitch Young on Forsyth Learner Profile, dress code, and goals alignment
- Founder, Co-President |** Physics Club at Denmark High School
Hosted 5+ Physics workshops, organized 3+ Physics competitions, earned Fermi Honor at Georgia Tech
- President, Officer |** Badminton Club at Denmark High School
- Captain |** JV Boys Tennis Team at Denmark High School
- Member |** Vertical Flight Society, Georgia Tech chapter

PROJECT PORTFOLIO

FRC Ball Intake + Shooter via CAN Bus	Commercial site stormwater modeling via HydroCAD
CO2 Dragster via Ansys CFD & FEA	Built Electric Rally Vehicle utilizing welding custom joints
Built ELRS FPV Racing Drone	Prototyped custom coin sorter utilizing laser engraver
Developed engineering drawings of residential annex	Prototyped automated water pump for water towers
Prototyped multi-tech dock via plasma cutter	Performed SimScale FEA on custom bike stand design